

The Architecture of the Nail

The hardened keratin you shape, and the living structure beneath it that makes the shaping possible

Learning Objectives

- Name and describe the structures of the nail unit and the function of each
- Explain where nail growth actually originates, and why that location matters
- State the approximate growth rate of fingernails and toenails, and why they differ
- Explain the composition of the nail plate and why it behaves differently from hair keratin

Before technique, understanding

The nail plate you file, buff, and polish is only the visible surface of a considerably more complex living unit. Every technique this course will teach you — where to push a cuticle, how hard to buff, when to decline a service — depends on understanding what lies beneath that hardened surface, and respecting that most of the nail's real biology is happening in places you cannot see or directly touch.

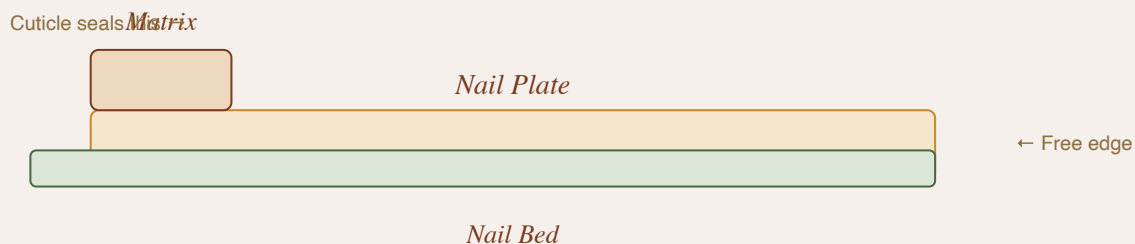
You are not decorating dead material floating free of the body. You are working at the visible end of a living structure that remembers how it was treated.

The nail unit, piece by piece

- **The nail plate** — the hard, visible nail itself, made of tightly layered dead keratinocytes.
- **The nail bed** — the skin directly beneath the plate, richly supplied with blood vessels, which is why healthy nail beds appear pink and why nail bed colour is a genuine, useful health indicator.

- **The matrix** — hidden beneath the base of the nail, under the cuticle, and the single most important structure in this module: this is where new nail cells are actually produced. Damage to the matrix, not the visible plate, is what causes permanent nail changes.
- **The lunula** — the pale, crescent-shaped visible portion of the matrix at the nail's base, sometimes hidden entirely under the cuticle depending on the individual.
- **The cuticle (eponychium)** — the thin layer of skin overlapping the base of the nail plate, sealing and protecting the matrix beneath it from bacteria and fungal invasion.
- **The hyponychium** — the skin beneath the free edge of the nail, forming a seal that keeps water and pathogens from entering underneath the nail plate.
- **The nail folds** — the skin framing the sides and base of the nail plate, holding it in place.

◆ Figure 1.1 · The Nail Unit



The matrix, hidden beneath the cuticle, is where the nail is actually made — the visible plate is simply where that history becomes visible.

Growth: slower than most clients assume

Fingernails grow at an average of about 3 millimetres per month, meaning a full fingernail regrows in roughly four to six months. Toenails grow considerably slower, around 1 millimetre per month, taking twelve to eighteen months to fully replace — a fact worth sharing with any client frustrated by how long a damaged toenail takes to "fix itself." Growth rate varies with age (faster in youth, slowing with age), circulation, and season (often faster in warmer months), but the underlying source never changes: all new growth originates at the matrix and pushes the older nail forward.

Hard keratin, not soft keratin

Recall from Course II that hair is built from keratin. Nails are built from keratin too — but a different, harder form of the same protein family, packed more densely and with a much lower moisture content than hair's softer keratin. This is precisely why nails are more brittle and prone to cracking under mechanical stress, rather than the stretch-and-snap failure typical of hair, and why nail health responds to hydration differently than hair does.

Key Terms

0

[object Object]

1

[object Object]

2

[object Object]

3

[object Object]

Practical Activity · Draw the Unit

Sketch a labelled cross-section of the nail unit — all seven structures from this module. Beside your drawing, write one sentence per structure in your own words. **Keep this drawing. It is the first page of your capstone portfolio in Module Fifteen.**